MIL-S-28851B(EC)
28 February 1983
SUPERSEDING
MIL-S-28851A
9 January 1981

MILITARY SPECIFICATION

SWITCHBOARD, TELETYPEWRITER SIGNAL, SECURE SB-3917/BGC AND RECEIVER AUDIO SB-3918/BRR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two 16 by 16 switching matrices. One is a secure teletype switching matrix that connects certain devices to teletypewriter and cryptographic equipment, as well as to several minicomputer input/output (I/O) channels, hereinafter referred to as Unit A. The other is a receiver audio switching unit that connects several receiver outputs to specified audio equipment, hereinafter referred to as Unit B.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-642	Plugs, Telephone, And Accessory Screws, General Specification For
MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification For
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories And Repair Parts, Packaging And Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-188-114	Military Characteristics Of Digital Interface Circuits

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

MIL-STD-189	Racks, Electrical Equipment, 19-Inch And Associated Panels
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Emission Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environment Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MIL-STD-1680	Installation Criteria For Shipboard Secure Electrical Information Processing Systems

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein.

NATIONAL SECURITY AGENCY

NACSEM 5100 A	Compromising Emanations Laboratory Test Standards (CELTS) Electromagnetics

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The teletypewriter signal and receiver audio switchboard shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.1.1 Modular construction. Modular construction shall be in accordance with Requirement 73 of MIL-STD-454.

3.1.1.2 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure 1.

3.1.1.3 Fungus nutrient parts or materials. Fungus nutrient parts of materials shall not be used in the equipment.

3.1.1.4 Description. Unit A shall have the capability of switching binary direct current signals of rates up to 2400 baud from each of 16 input devices to 16 output devices. Unit B shall have the capability of switching audio signals in the range of 300 hertz (Hz) to 3500 Hz and a digitized audio signal of 25 kilohertz (kHz) bandwidth from each of 16 inputs to 16 output devices. The switching in each unit shall be performed by thumbwheel switches mounted on the front panel of the enclosure. The input equipment connection status shall be indicated by numbers displayed on switch cylinder dial through a fixed window.

3.1.1.5 Front panel diagrams. FIGURE 1 shows the front panel layout of both Units A and B. The information that shall be imprinted on the input equipment list card attached to the front panels of Units A and B is specified in 3.1.1.6. The information that shall be imprinted on the output equipment (switch) legend attached to the front panels of Units A and B is specified in 3.1.1.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

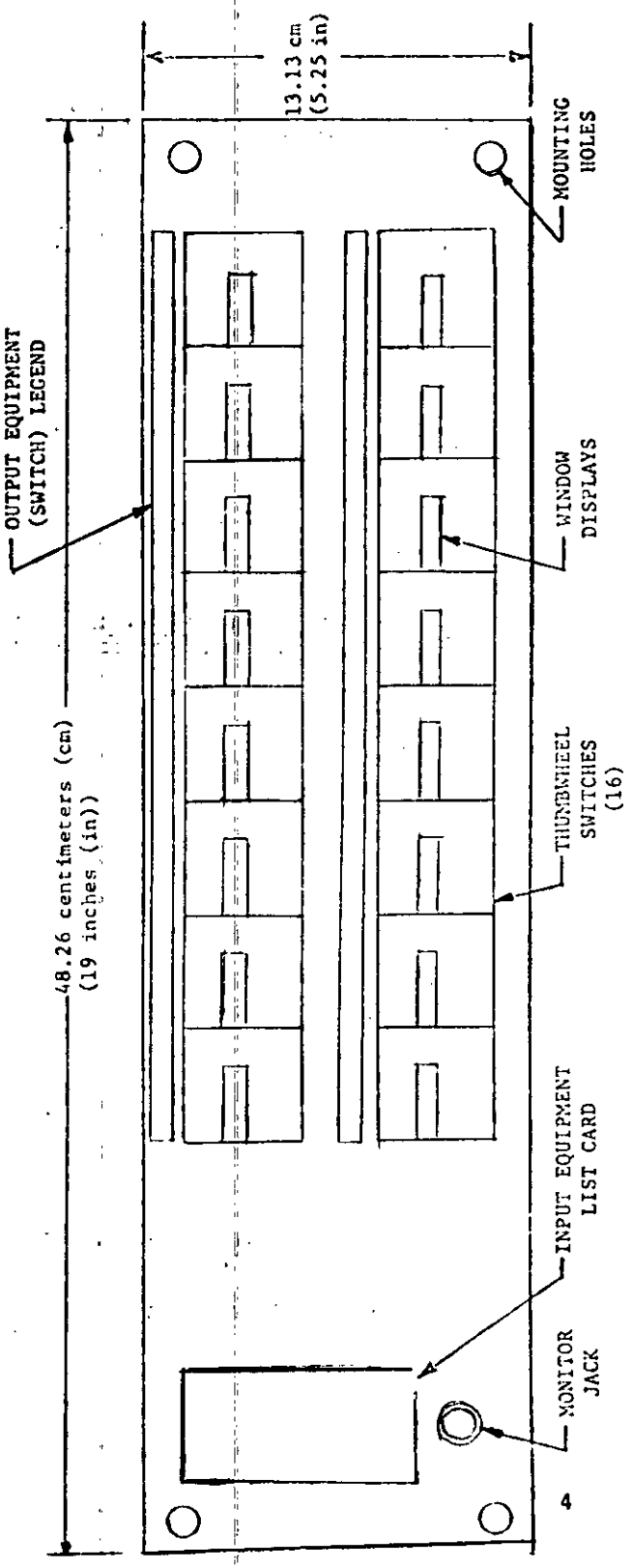


FIGURE 1. Front panel for Units A and B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

3.1.1.6 Input equipment list card nomenclature. The input equipment list cards for Units A and B shall be labeled as follows:

Unit A		Unit B	
T/W switch position	Input equipment	T/W switch position	Input equipment
1	KYBD #1	1	1051#1 USB
2	KYBD #2	2	1051#1 LSB
3	KYBD #3	3	1051#2 USB
4	KYDB #4	4	1051#2 LSB
5	TT-603	5	WRR-3
6	SIU PNTR	6	R-1738
7	SIU SSIXS	7	WSC-3#1
8	WRR-7 CH A	8	WSC-3#2
9	WRR-7 CH B	9	SPARE
10	WRR-7 CH C	10	SPARE
11	WRR-7 CH D	11	SPARE
12	KWR-37	12	SPARE
13	KW-7(R)	13	SPARE
14	ON-143 #1(R)	14	SPARE
15	ON-143 #2(R)	15	SPARE
16	SPARE/OFF	16	SPARE/OFF

3.1.1.7 Output equipment (switch) legend nomenclature. The output equipment (switch) legend for Units A and B shall appear on the front panel and be labeled as follows:

Unit A		Unit B	
Switch	Label	Switch	Label
1	PNTR #1	1	SPKR #1
2	PNTR #2	2	SPKR #2
3	PNTR #3	3	SPKR #3
4	PNTR #4	4	SPKR #4
5	TT 605 #1	5	URA-17 CH A
6	TT 605 #2	6	URA-17 CH B
7	SIU KYBD	7	CV-2460
8	SIU SSIXS	8	KY-8
9	SIU VERDIN	9	KY-75
10	KWR-37(I)	10	NAV CTR
11	KW-7(T)	11	USQ-76 USB
12	ON 143 #1(T)	12	USQ-76 LSB
13	ON 143 #2(T)	13	RHMS
14	MONITOR	14	MONITOR
15	SPARE	15	SPARE
16	SPARE	16	SPARE

3.1.2 Crosspoint diagrams. FIGURE 2 shows the crosspoints that are required to be actuated for Unit A. FIGURE 3 shows the crosspoints that are required to be actuated for Unit B.

3.1.3 Switch characteristics. Thumbwheel switches used for both Units A and B shall be such that switch actuation will provide separate contacts for both the signal lines and the signal return lines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

SWITCH #1	OUTPUT[1] CONN #	OUTPUT EQUIPMENT	INPUT EQUIPMENT CONNECTION FOR SWITCH POSITION															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	J17	PNTR #1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2	J18	PNTR #2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
3	J19	PNTR #3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4	J20	PNTR #4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
5	J21	TT 605 #1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
6	J22	TT 605 #2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
7	J23	SIU KYBD	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
8	J24	SIU SSIXS	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
9	J25	SIU VERDIN	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
10	J26	KWR-37(T)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
11	J27	KW-7 (T)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
12	J28	ON 143 #1(T)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
13	J29	ON 143 #2(T)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
14	J30	MONITOR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
15	J31	SPARE	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
16	J32	SPARE	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

INPUT EQUIPMENT NUMBERS		INPUT[1]
SWITCH	WINDOW DISPLAY	CONN
1	1-16	J1
2	1-16	J2
3	1-16	J3
4	1-16	J4
5	1-16	J5
6	1-16	J6
7	1-16	J7
8	1-16	J8
9	1-16	J9
10	1-16	J10
11	1-16	J11
12	1-16	J12
13	1-16	J13
14	1-16	J14
15	1-16	J15
16	1-16	J16

NOTE: [1]Rear panel connectors.

FIGURE 2. SB-3917 crosspoint interconnection diagram.

Source: https://assistdla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

SWITCH #1	OUTPUT[1] CONN #	OUTPUT EQUIPMENT	INPUT EQUIPMENT CONNECTION FOR SWITCH POSITION															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	J17	SPKR #1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2	J18	SPKR #2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
3	J19	SPKR #3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4	J20	SPKR #4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
5	J21	URA-17 CH A	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
6	J22	URA-17 CH B	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
7	J23	CV-2460	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
8	J24	KY-8	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
9	J25	KY-75	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
10	J26	NAV CTR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
11	J27	USQ-76 USB	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
12	J28	USQ-76 LSB	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
13	J29	RHMS	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
14	J30	MONITOR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
15	J31	SPARE	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
16	J32	SPARE	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

INPUT EQUIPMENT		INPUT[1]
SWITCH	WINDOW DISPLAY	CONN
1	1-16	J1
2	1-16	J2
3	1-16	J3
4	1-16	J4
5	1-16	J5
6	1-16	J6
7	1-16	J7
8	1-16	J8
9	1-16	J9
10	1-16	J10
11	1-16	J11
12	1-16	J12
13	1-16	J13
14	1-16	J14
15	1-16	J15
16	1-16	J16

NOTE: [1]Rear panel connectors.

FIGURE 3. SB-3918 crosspoint interconnection diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

3.1.4 Unit A interface requirements. Unit A shall be designed to operate in accordance with the low-level input and output (positive and negative 6 ±1 volt (V)) standard interface requirements of MIL-STD-188-114. A summary of pertinent characteristics is included in 3.1.4.1 and 3.1.4.2.

3.1.4.1 Input equipment impedance. The output impedance of an input equipment, used in low-level circuits, does not exceed 100 ohms for currents whose magnitude is less than 0.01 ampere (A).

3.1.4.2 Output equipment resistance. The minimum input resistance of an output equipment, used in low-level circuits, is 6 kilohms (kohms). The maximum total input capacitance of the device shall be 2500 picofarads.

3.1.5 Unit B interface requirements. The switchboard crosspoints shall be capable of carrying the following types of signals:

a. 600-ohm balanced audio signals from 300 Hz to 3500 Hz, up to 20 milliwatts (mW)

b. 600-ohm unbalanced digitized audio signals of 25 kHz bandwidth, up to 20 mW

3.1.5.1 Terminal impedance. The switchboard shall accommodate I/O devices having terminal impedance of 600 ohms in the frequency range between 300 Hz and 3500 Hz.

3.1.5.2 I/O power levels. The nominal power levels available at the I/O terminals of the switchboard shall be 0 decibels (db) referred to one milliwatt (dBm).

3.2 Characteristics.

3.2.1 Equipment selection. Selection of I/O device conditions shall be made by use of thumbwheel switches located on the front panel of each unit. Each switch shall represent an output device, and the positions on a switch shall represent the input devices that can be connected to that output device.

3.2.2 Display nomenclature. The number of the input device selected by a thumbwheel switch on either unit shall appear through the display window. The number in the display window for a specific equipment connection is specified in 3.1.1.6.

3.2.3 Units A and B monitor jacks. Units A and B shall each have a front panel local monitor jack to permit the monitoring of any of the input devices. This jack shall be labeled LOC MON.

3.2.4 Internal power supply requirements. With low-level switching, no internal power supply is required to produce the keying voltage for the output switch. This voltage is derived from the input device.

3.2.5 Transmission performance requirements. Each unit shall conform to the transmission characteristics specified in 3.2.5.1 and 3.2.5.2.

3.2.5.1 Unit A transmission performance requirements. For rates up to 2400 baud, Unit A shall conform to, or exceed, the following transmission characteristics:

a. Insertion loss through the switch shall not exceed 0.5 dB.

b. Crosstalk isolation shall be 60 dB minimum.

c. Noise on any connection through the switch, when idle and properly terminated, caused by the switch itself, shall not exceed -69 dBm in a 3200-Hz bandwidth between 300 Hz and 3500 Hz. Impulse noise on any connection, when idle and properly terminated, due to switching on another connection or due to any other switch function not associated with the connection measured, shall not exceed a 4-millivolt (mV) peak when measured across a 600-ohm load.

d. Balance between the marking and spacing voltages shall be within one percent of each other when a symmetrical binary signal is passed through the switch. For this specification, balance shall be:

$$\frac{V_{pos} - V_{neg}}{V_{pos} + V_{neg}} \times 100$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

3.2.5.2 Unit B transmission performance requirements. Unit B shall conform to the following transmission characteristics:

a. Insertion loss through the switch shall not exceed 0.5 dB when measured with a 1-kHz test tone.

b. Crosstalk isolation shall be 60 dB minimum for frequencies between 300 Hz and 3500 Hz and shall be 40 dB minimum for frequencies between 3500 Hz and 25 kHz.

c. Harmonic distortion through any switch connection shall be at least 40 dB below the fundamental for frequencies between 300 Hz and 25 kHz.

d. Intermodulation distortion through any switch connection shall be at least 40 dB below the fundamental for frequencies between 300 Hz and 25 kHz.

e. Delay distortion through any switch connection shall not exceed 30 microseconds in the frequency band between 300 Hz and 25 kHz.

f. Maximum signal power that shall be passed by a switch connection without limiting or overloading shall be at least 20 mW (+13 dBm) for frequencies between 300 Hz and 25 kHz.

g. Longitudinal balance of any connection through the switch shall be at least 40 dB for frequencies between 300 Hz and 3500 Hz.

h. Attenuation of all frequencies between 300 Hz and 3500 Hz shall be within the limits of ±1.0 dB with respect to the attenuation at 1000 Hz, and the attenuation of all frequencies greater than 3500 Hz and less than 25 kHz shall be within the limits of ±1.5 dB with respect to the attenuation at 1 kHz.

i. Noise on any connection through the switch, when idle and properly terminated, caused by the switch itself, shall not exceed -60 dBm in a bandwidth between 300 Hz and 25 kHz. Impulse noise on any connection, when idle and properly terminated, due to switching on another connection, or due to any other switch functions not associated with the connection, shall not exceed a 4-mV peak amplitude when measured across a 600-ohm load over 1 megahertz bandwidth.

j. Input and output impedance of the switch in the frequency band between 300 Hz and 25 kHz shall have a return loss of at least 20 dB when referred to a 600-ohm resistance.

3.2.6 Security criteria. Unit A shall conform to the applicable criteria for secure electrical processing systems as specified in MIL-STD-1680 and NACSEM 5100 A.

3.2.7 Physical characteristics. Both Units A and B shall be housed in enclosures with the maximum dimensions and weight shown in TABLE I.

TABLE I. Dimensions and weight.

Width		Depth		Height		Weight	
cm 1/	(in.) 2/	cm	(in.)	cm	(in.)	kilograms	(pounds)
48.26	(19)	50.8	(20) (including rear connectors)	13.34	(5.25)	9	(21)

1/ Centimeters
2/ Inches

3.2.7.1 Mounting. Units A and B, with enclosures, shall be capable of being mounted in a standard 48.26-cm (19-in.) electrical equipment rack as specified in MIL-STD-189.

3.2.7.2 Connectors. Rear panel connectors for both Units A and B shall be as follows:

Input connectors J1 through J16: MS3402-10SL-3P
Output connectors J17 through J32: MS3402-10SL-3S

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

For both input and output connectors pin connections shall be as follows:

Pin A: Signal
Pin B: Signal return
Pin C: Shield

Front panel monitoring jacks shall be as follows:

Unit A: MT-342B, MIL-P-642(A)
Unit B: NSN 5935-000-237-2494

3.2.7.3 Enclosures. The enclosures shall be dripproof (45 degrees) as specified in the Dripproof paragraph of MIL-STD-108.

3.2.7.4 Front panel. The front of each unit shall be constructed to permit easy access to the rear of the components mounted on the panel and to components located within the enclosure. The design of controls and indicators, and their arrangement on operator and maintenance panels, shall conform to the criteria of MIL-STD-1472.

3.2.7.5 Enclosure bonding. The enclosure for each unit shall be furnished with mounting holes for bonding-strap ground connectors.

3.2.7.6 Exterior finish. The exterior finish of both units shall conform to the requirements of MIL-E-15090; however, the requirements of MIL-STD-1680 shall take precedence in the case of Unit A.

3.2.7.7 Connectors. The connectors used in both units shall conform to Class D of MIL-C-5015.

3.2.7.8 Cables. Cables for the I/O devices are not required; however, mating connectors for the I/O device connectors on each unit shall be provided.

3.2.7.9 Conformal coating. To facilitate I/O equipment circuit reconfiguration, conformal coating is not required.

3.2.8 Reliability.

3.2.8.1 Quantitative reliability requirements. The upper test mean-time-between-failures (MTBF) (θ_0 as defined by MIL-STD-781) shall be 2200 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1100 hours.

3.2.8.2 Component reliability. All switches shall have a minimum life of 100,000 operations.

3.2.9 Maintainability.

3.2.9.1 Quantitative corrective maintenance. Units A and B shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 0.5 hour and a maximum-corrective-maintenance-time (M_{maxct}), 95th percentile, not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level by the replacement of lowest subassemblies (modules, printed circuit boards) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, alignment, and checkout for all corrective maintenance actions. The equipment is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.2.9.2 Subassembly (module) maintainability. Repairable switch modules shall have a M_{ct} of 2.0 hours and a M_{maxct} of 4.0 hours (95th percentile) at the organizational, intermediate, or depot level, as appropriate. M_{ct} for module repair includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly printed circuit card for which repair is accomplished by the replacement of piece parts.

3.2.10 Environmental service conditions. The equipment shall be capable of continuous reliable operation, within the performance limits specified herein, under the environmental conditions specified in 3.2.10.1 through 3.2.10.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

3.2.10.1 Temperature. The operating and nonoperating environment temperature shall be classified as Range 4 except the operating temperature limits shall be -28° Celsius (C) to 65°C.

3.2.10.2 Vibration. The equipment shall withstand the Type I vibration test of MIL-STD-167-1 at an upper vibration frequency limit of 55 Hz.

3.2.11 Electromagnetic compatibility. Units A and B shall conform to the requirements of MIL-STD-461, Class A5, test category telephone switchboard except as follows:

a. CS02 and CS09 are required.

b. RS03 shall be performed at 10 volts per meter.

3.3 Output-device name tags. The output-device name tags that appear above each switch on the front panel of each unit shall be attached so as to permit easy replacement.

3.4 Input-equipment list. The input-equipment list card on the front panel of each unit, listing the input equipment, shall be attached so as to permit easy replacement.

3.5 Unacceptable materials. Equipment design shall not include polychlorinated-biphenyls, asbestos and asbestos compounds, fragile or brittle materials, beryllium and beryllium compounds unless so identified, and lithium and lithium compounds not specifically approved by the procuring activity.

3.6 Printed circuit assembly protection. When the equipment is in its normal operating mode, removal or insertion of printed circuit assemblies shall not damage the assembly or any other part of the equipment. Each assembly shall be keyed to prevent incorrect insertion.

3.7 Human engineering. Equipment design shall conform to the requirements of MIL-STD-1472.

3.8 Logistics.

3.8.1 Maintenance. Units A and B shall use plug-in assemblies wherever feasible in order to reduce maintenance and repair effort.

3.8.1.1 Special maintenance items. Special tools or items required for maintenance shall be provided (see 6.5), as part of the equipment, by the contractor.

3.9 Workmanship screen. Units A and B shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz (see 4.7.2.1). Units A and B shall withstand a temperature cycling screen (see 4.7.2.2) to assure elimination of initial failures due to random defective components and marginal operation due to component lot variations or poor assembly processes and procedures.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection

1. In-process inspection (see 4.4.1)

2. Production inspection (Group A) (see 4.4.2)

3. Production control inspection (Group B) (see 4.4.3)

4. Environmental inspection (Group C) (see 4.4.4)

c. Reliability testing (see 4.6.4)

d. Maintainability demonstration (see 4.5.5)

e. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2), one unit each of the SB-3917/BGC and the SB-3918/BRR shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

4.4 Quality conformance inspection.

4.4.1 In-process inspections. Each equipment produced shall be subjected to the inspections of 4.7 through 4.7.2.2.

4.4.2 Production inspection (Group A). Production inspection shall be made on every item offered for delivery. This inspection shall comprise such examinations and tests which will provide the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests listed in Group A of TABLE II.

4.4.3 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified herein. This inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of equipment to withstand this heat; tests of efficiency; and tests of the performance with other equipment in the system. These tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests shown in Group B of TABLE II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

TABLE II. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	In-process inspection	Quality conformance inspection		
					Group A	Group B	Group C
Surface examination							
Weight	3.2.7	4.6	X			X	
Size	3.2.7	4.6	X		X	X	
Parts control	3.1.1.2	4.6	X		X		
Finish	3.2.7.7	4.6	X		X		
Marking	3.1	4.6	X		X		
Safety	3.1	4.6	X		X		
Workmanship	3.1	4.7	X		X		
Pre-performance		4.6	X			X	X
Performance	3.1, 3.2	4.6	X		X		
Proper crosspoints switching	3.1.2	4.6	X		X		
Impedance and resistance	3.1.4.1, 3.1.4.2, 3.1.5.1	4.6	X		X		
Transmission	3.2.5.1, 3.2.5.2	4.6	X				
Display readability	3.2.2	4.6	X		X		
Insertion loss	3.2.5.1a, 3.2.5.2a	4.6	X		X		
Crosstalk isolation	3.2.5.1b, 3.2.5.2b	4.6	X		X		
Signal power passed by switch	3.2.5.2f	4.6	X		X		
Temperature							
Low temperature	3.2.10.1	4.6	X				X
High temperature	3.2.10.1	4.6	X				X
Humidity	3.1	4.6	X				X
Fungus	3.1	4.6	X				
Workmanship screen	3.9	4.7.2	X		X		
Interchangeability	3.1	4.6.1	X				X
Thermal survey	3.1	4.6	X				X
Shock	3.1	4.6	X				X
Salt fog (spray)	3.1	4.6	X				
Vibration	3.2.10.2	4.6	X				X
Inclination	3.1	4.6	X				X
Enclosures	3.2.7.3	4.6	X			X	
Electromagnetic interference	3.2.11	4.6.4	X				X
Reliability and test demonstration	3.2.8	4.6.2	X				
Maintainability demonstration	3.2.9	4.6.3	X				
General workmanship	3.1	4.7		X	X		
Workmanship screen	3.9	4.7.2		X	X		
Vibration	3.9	4.7.2.1		X	X		
Temperature cycling	3.9	4.7.2.2		X	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

4.4.3.1 Sampling plan for Group B inspection. Equipment submitted for Group B sampling tests shall be selected by the Government Quality Assurance Representative (QAR) and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level shall be 6.5 percent defective (based on all tests performed on the lot) and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.3.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.4 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified herein. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests, such as extremes of temperature and humidity, and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests shown in Group C or TABLE II.

4.4.4.1 Sampling for environmental inspection. One equipment shall be selected from each successive 50 equipments, or fraction thereof, produced. The first sample equipment shall be selected from the first month's production.

4.4.4.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.4, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The equipment shall then be retested.

4.4.5 Reinspection of conforming production control and environmental test units. Unless otherwise specified (see 6.2), sample units which have been subjected to and have passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.5 Test equipment and test conditions. Test equipment and test conditions for first article inspection and quality conformance inspection shall comply with the following quality assurance provisions of MIL-E-16400.

a. Test facilities, chambers, and apparatus

b. Standard test conditions

c. Performance of test

4.6 Test methods.

4.6.1 Inspection for interchangeability. The requirements for interchangeability specified in MIL-E-16400 shall be demonstrated on two switchboards chosen at random from each lot of 50 or less.

4.6.2 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781, Test Plan IV C. Test temperature shall be maintained at 50°C. Vibration stress (continuous) shall be specified in the Combined environments for shipboard equipment paragraph of MIL-STD-781.

Unit A.	Source impedance:	100 ohms
	Input voltage level:	±(6±1)V
	Load:	6 kohms
Unit B.	Source impedance:	600 ohms, balanced and unbalanced
	Input power level:	0 dbm (600 ohm)
	Load:	600 ohm

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

4.6.3 Maintainability demonstration.

4.6.3.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment M_{ct} requirements are fulfilled.

4.6.3.2 Demonstrated conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual printed circuit boards (PCB)) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Sixty candidate corrective maintenance tasks shall be identified (selected in accordance with Appendix A of MIL-STD-471). The following information shall be made available for each candidate task:

a. Designation of specific faulty part

b. Failure mode

c. Means of introducing fault (substitution of faulty part or simulation thereof)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.6.3.3 Subassembly (module) maintainability. The module maintainability requirements shall be demonstrated by replacement of piece parts at the organizational, intermediate, or depot-level, as appropriate. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. A total of 20 faults shall be selected by the procuring activity for the module maintainability demonstration.

4.6.3.4 Accept or reject criteria.

4.6.3.4.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as shown in TABLE III.

TABLE III. Acceptance criteria.

	M_{ct}	M_{maxct}
Acceptance level	8	0
Sample size	20	20

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified in index (M_{ct}, M_{maxct}) is less than or equal to that shown in TABLE III corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept detection can only be made when an accept decision is made for both M_{ct} and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.6.3.5 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the equipment shall not interfere with on-line operation.

4.6.3.6 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.6.3.7 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.6.3.8 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

4.6.3.9 Rejection. Failure (see 6.4) to conform to any of the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies

b. Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.6.3.10 Acceptance. The maintainability demonstration shall exhibit that the equipment conforms to the accept criteria prior to any delivery.

4.6.4 Electromagnetic compatibility test. The requirements of 3.2.11 shall be verified by tests performed in accordance with MIL-STD-462. For Group C, only test CE03 and RS03 shall be performed.

4.7 Workmanship.

4.7.1 General workmanship. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process and at the end item level for conformance to the requirements of 3.1.

4.7.2 Workmanship screen. The workmanship screen shall consist of random vibration and temperature cycling. The screening may be performed at the module, drawer, or end item level. The random vibration shall be performed on each equipment, including cables and connectors, prior to conducting the temperature cycling test (see 4.7.2.2).

4.7.2.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 4. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD).

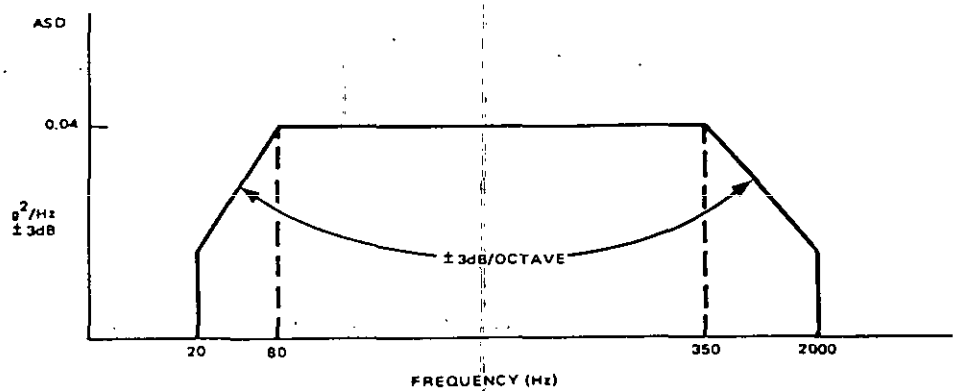


FIGURE 4. Random vibration curve.

4.7.2.2 Temperature cycling. Each equipment shall be subjected to 10 cycles of the temperature curve shown in FIGURE 5. The temperature rate of change shall be not less than 5°C per minute. Equipment power shall be turned on and off at the indicated times. When practical, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition. When failures occur, the equipment shall be reworked and the cycling continued for a cumulative total of 10 cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

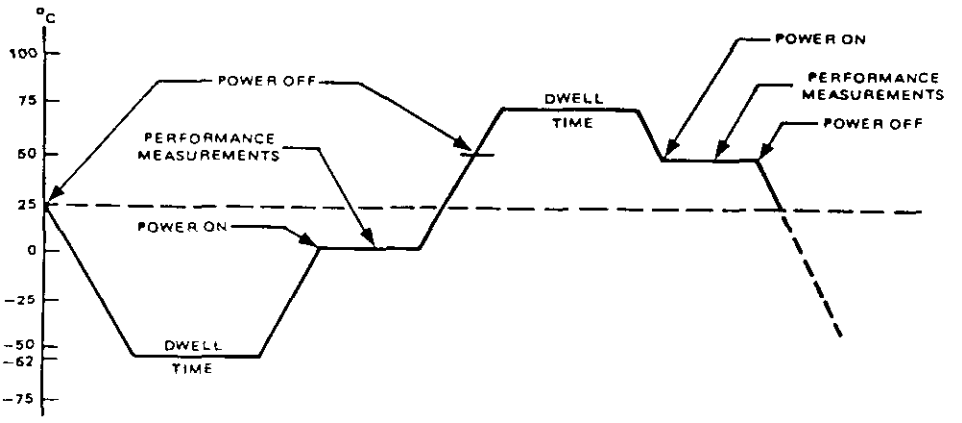


FIGURE 5. Temperature curve (one cycle).

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for use in submarines.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Number of first article samples to be submitted if other than specified in 4.3

c. When reinspected production control and environmental sample units may not be accepted (see 4.4.5)

d. Levels of preservation, packaging, packing, and marking required (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulations (DAR) 7-1049 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

Paragraph	Data Requirements	Applicable DID
4.4	Procedures, Test	UDI-T-22710
4.4	Test Report	DI-T-5247
4.4	Report, Failure	DI-R-7039
4.6.2	Reports, Reliability Test and Demonstration	DI-R-7034
4.6.2	Procedures, Reliability Tests	DI-R-7035
4.6.3	Plan, Maintainability Demonstration	DI-R-2129
4.6.4	Report, Test, Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC)	DI-R-2066/R021
4.6.3	Report, Maintainability Demonstration	DI-R-2130

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first articles should be first production items. The first article should consist of one each of Units A and B. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Failure. A failure shall be any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements specified herein.

6.5 Special tools. Special tools are defined as those not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY - EC
(Project 5895-N304)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 02 MAR

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:51Z
Check the source to verify that this is the current version before use.